# Muhammad Mustafa Shah

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## Profile & Research Interests

**Research Interests:** Computer Vision & Perception, Multimodal Deep Learning, Edge AI, and Autonomous Systems.

Machine learning engineer and Data Science graduate working at the intersection of computer vision, multimodal fusion, and latency-constrained deployment. Engineered detection-and-localization models that run at 700+ FPS with 71% fewer parameters than CNN baselines, compressed perception pipelines to 18 ms CPU latency via INT8 quantization, and applied deep learning to cross-disciplinary multimodal data. Seeking an MASc position to extend this work to perception, prediction, and decision-making for autonomous vehicles.

## Education

**Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)** **Aug 2022 – June 2026**
*Bachelor of Science in Data Science | CGPA: 3.24/4.00 (Dean's Honour List)* *Topi, Pakistan*

- Final four semester GPAs: 3.50, 3.61, 3.73, 3.73 (67 credits, weighted 3.64). Grade A in both parts of the Senior Design Project.
- **Statistics & Quantitative Methods:** Inferential Statistics & Applied Probability, Linear Algebra & Optimization, Multivariate Calculus, Differential Equations, Design & Analysis of Algorithms, Parallel Processing.
- **Machine Learning:** Deep Learning, Computer Vision & Image Processing, Data Mining, Big Data Analytics, Generative AI with Large Language Models.

## Research & Professional Experience

**Osol Tech (Pvt.) Ltd, Fatima Group** **June 2026 – Present**
*AI Trainee Engineer* *Pakistan*

- Develop, test, and optimize deep learning models and end-to-end AI pipelines for production-grade vision and speech systems.
- Validate that offline benchmark gains translate to measured latency and accuracy on target hardware, bridging the gap between theoretical metrics and real-time edge deployment.
- Adapt large pretrained models to narrow domains using parameter-efficient methods, ensuring scalable integration into high-throughput autonomous environments.

**BIOMISA Lab, NUST** **July 2025 – Sep 2025**
*Research Assistant, Multimodal AI & Deep Learning* *Islamabad, Pakistan*

- Applied deep learning to cross-disciplinary domains, engineering multimodal Vision-Language Models (VLMs) and processing 50+ complex biomedical imaging and signal datasets.
- Optimized backend feature-extraction pipelines, reducing model inference latency by 40% to enable near-real-time model queries.
- Ran controlled architectural comparisons under fixed evaluation protocols to select the most efficient deployable candidates, ensuring 95% data integrity on secure dashboards.

**ITSOLERA Pvt Ltd** **June 2024 – Aug 2024**
*Machine Learning Intern (Data Science)* *Remote*

- Fine-tuned transformer-based (RoBERTa) models for classification tasks, establishing NLP applications with an 85% F1-score and transitioning models into reproducible Python pipelines.
- Built scalable ETL workflows processing 100GB+ of streaming data in Python and SQL to support high-throughput data ingestion.
- Deployed trained inference endpoints on dashboard infrastructure, optimizing serving pipelines to reduce query response time to under 1 second.

## Research Projects (Computer Vision & Multimodal Learning)

**Interpretable Vision-Language Framework (Multimodal Learning)** | *PyTorch, VLMs, SHAP*

- Developed a cross-disciplinary multimodal pipeline combining complex imaging with unstructured textual data using Vision-Language Models (VLMs), improving predictive accuracy by 12% over unimodal baselines.
- Embedded Explainable AI (XAI) mechanisms using attention-rollout and SHAP to generate transparent rationales, demonstrating trustworthy deep learning for high-stakes perception tasks.

**Real-Time Visual Detection & Spatial Localization on Edge Hardware** | *PyTorch, MobileNetV2, CNNs*

- Engineered a hybrid MobileNetV2 vision detector achieving 84.81% accuracy and sustaining 711 FPS for real-time spatial localization on edge-class hardware.
- Reduced trainable parameters by 71% (1.96M) compared to deep-CNN baselines via a circuit-inspired residual block, optimizing perception architectures for resource-constrained autonomous nodes.
- Implemented a context-aware ROI-crop pipeline that increased effective object area from ∼0.1% to ∼11% of the input, enabling small-object bounding-box regression at 0.73 mean IoU and an 89.6% detection rate.

**Low-Latency Perception via Model Compression & INT8 Quantization** | *YOLO, MobileNetV2, TinyML*

- Replaced a YOLO backbone with depthwise-separable MobileNetV2 and applied post-training INT8 quantization, cutting CPU inference latency from 145 ms to 18 ms (55.5 FPS).
- Compressed model size by 92% (28.0 MB to 2.2 MB) for only a 4% precision drop, proving viability for GPU-free, real-time object detection on embedded vehicular compute modules.

**Multimodal Agentic Fusion Pipeline for Streaming Sensor Data** | *LangGraph, LLMs, Kafka*

- Architected a decentralized multimodal system as a Directed Acyclic Graph of specialized agents (vision, speech, text) to asynchronously process 50+ hours of streaming input daily.
- Enforced system trustworthiness with Retrieval-Augmented Generation (RAG) and semantic grounding to suppress hallucinations during cross-modal reasoning and synthesis.

**Parameter-Efficient Multimodal Adaptation under Distribution Shift** | *PyTorch, LoRA/PEFT*

- Fine-tuned a 520M-parameter multimodal model to handle targeted domain shifts using 210 low-rank adapters, training just 0.1% of weights with the base model frozen.
- Reversed a systematic failure in baseline representations (correlation shifted from $-0.164$ to $+0.093$) and merged adapters back into base weights for zero added inference cost.

**Real-Time Temporal Anomaly Detection for Streaming Telemetry** | *PyTorch, LSTMs, Kafka*

- Devised a predictive LSTM engine to detect sudden temporal anomalies and concept-drift in continuous, non-linear sequential sensor data, improving predictive accuracy by 15%.
- Designed the inference path to process 5 years of streaming data continuously via distributed Kafka pipelines, bypassing batch-processing bottlenecks for real-time monitoring.

## Technical Skills

**Deep Learning & Vision**: CNNs, Object Detection & Spatial Localization (MobileNetV2, YOLO), Semantic Feature Fusion, Vision–Language / Multimodal Models, Transformers, Deep Reinforcement Learning Concepts, Attention-Based Interpretability (SHAP).
**Efficient & Edge AI**: TinyML, Post-Training INT8 Quantization, Model Compression, Depthwise-Separable Architectures, Latency/FPS Profiling, Real-Time Edge Inference on resource-constrained hardware.
**Programming & Frameworks**: Python (PyTorch, OpenCV, Scikit-learn, Pandas, NumPy), C/C++, FastAPI, LangGraph, Git.
**Infrastructure & Distributed Systems**: Docker, Linux, Apache Kafka, PySpark, SQL, AWS (EC2), DAG-based orchestration, asynchronous message-queue coordination.
**Statistical & Experimental Method**: Controlled benchmarking under fixed evaluation protocols, error-distribution analysis, failure-mode profiling.
**Languages**: English (fluent, medium of instruction throughout degree), Urdu, Pashto (native).

## Technical Reports

M. M. Shah, "Enhancing Voice Cloning for South Asian English Accents via LoRA Fine-Tuning of Chatterbox TTS," 2026.
M. M. Shah, "Efficient PCB Defect Detection via CURA-Based Lightweight Vision Models," 2025.
*No peer-reviewed publications to date.*

## Honours, Awards & Leadership

Dean's Honour List and Competitive Merit Scholarship, GIKI **2022–2026**

Regional Participant, International Collegiate Programming Contest (ICPC), C++ **2024**

Best Dedication Award, Project Topi, GIKI Welfare Society **2024**

**Leadership:** Society Head, Project Topi, GIKI Welfare Society (2022–Present); Fellow & Project Lead, Amal Academy Fellowship (2025).